

Developing intervertebral discs degeneration grade classification according to the Pfirrmann scale

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1 Introduction

Intervertebral disc degeneration (IDD) is a major contributor to chronic low back pain. Magnetic resonance imaging (MRI) serves as the gold standard for IDD assessment, yet manual grading is often subjective and inconsistent. With advances in artificial intelligence (AI), particularly deep learning, automated detection and classification of IDD from MRI has become increasingly feasible (1).

Let's begin by introducing Pfirrmann grading. Information below comes from radiopedia.org, a peer-reviewed online encyclopedia for radiology (2). The Pfirrmann grading system is the most widely known classification for intervertebral disc degeneration and is used in both clinical and research contexts. The intra- and inter-observer agreement of the Pfirrmann grading system is excellent. Although is primarily intended for use in the lumbosacral region, it has been employed in the thoracic and thoracic regions also. Modified classification was proposed in 2007 by Grifth et al. (4) to better categorize degenerative discs in older patients where most discs are Pfirrmann grade III or IV. It has, however, not been as widely adopted.

1.1 Classes of degradation

The Pfirrmann grading system was first described in 2001 and focuses on the appearance of discs on sagittal T2-weighted scans.

1. grade I

- (a) disc is homogeneous with bright hyperintense white signal intensity and normal disc height

2. grade II

- (a) disc is inhomogeneous but retains the hyperintense white signal

- (b) nucleus and annulus are clearly differentiated, and a gray horizontal band could be present
- (c) disc height is normal

3. grade III

- (a) disc is inhomogeneous with an intermittent gray signal intensity
- (b) distinction between nucleus and annulus is unclear
- (c) disc height is normal or slightly decreased

4. grade IV

- (a) disc is inhomogeneous with a hypointense dark gray signal intensity
- (b) there is no more distinction between the nucleus and annulus
- (c) disc height is slightly or moderately decreased

5. grade V

- (a) disc is inhomogeneous with a hypointense black signal intensity
- (b) there is no more difference between the nucleus and annulus
- (c) the disc space is collapsed

1.2 Drawbacks of Pfirrmann classification

Pfirrmann grading has been proven valuable in both clinical and research settings due to its simplicity and ability to standardize the evaluation of intervertebral disc degeneration (IDD) (2). However, despite its widespread use, the Pfirrmann system also possesses several limitations. One of the primary drawbacks is its reliance on subjective visual assessment, which can lead to significant inter- and intra-observer variability. Several studies have shown that different radiologists may interpret the same MRI images differently, resulting

in inconsistent grading outcomes (3). This inconsistency complicates the comparison of diagnostic results across studies or clinical practices. Moreover, the Pfirrmann system does not take into account other critical factors such as Modic changes or the presence of high-intensity zones, which have been shown to correlate with more advanced disc degeneration and low back pain (2). These limitations highlight the need for more objective and comprehensive grading systems that can integrate additional imaging biomarkers beyond disc morphology. Some modifications to the Pfirrmann system have been proposed, but variability and the limited scope of these changes remain challenges in the consistent assessment of IDD (4).

2 Dataset

The SPIDER dataset consists of 447 sagittal T1- and T2-weighted MRI series obtained from 218 studies, corresponding to 218 patients with a history of low back pain. For each MRI image, segmentation masks of the intervertebral discs are provided. In addition, radiological assessments of Pfirrmann grades for each intervertebral disc level are included in a separate file. This dataset allows for the analysis and classification of disc degeneration on a per-disc basis while providing ground truth information for both anatomical localization and Pfirrmann grading. Based on the provided file containing the ground truth, we calculated the distribution of Pfirrmann grades across the dataset. The analysis revealed that grades 1 and 5 were the least represented, while grade 3 had the highest number of samples 1. This indicates an imbalanced distribution of disc degeneration levels within the dataset.

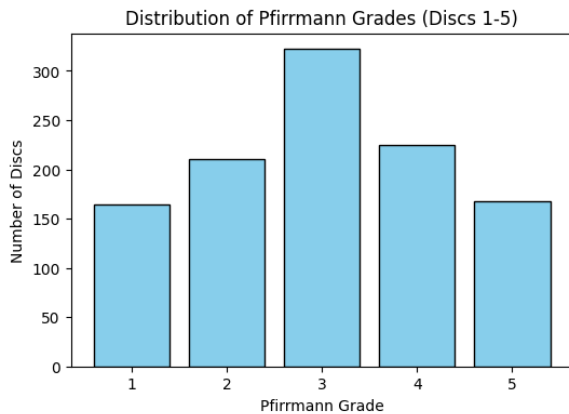


Figure 1: Distribution of Pfirrmann grades across SPIDER dataset.

3 Data preprocessing

All preprocessing steps were applied systematically to ensure consistency and prepare the dataset for model training. First, only T2-weighted images were selected. Next, all images and their corresponding masks were reoriented to the LPS (Left-Posterior-Superior) coordinate system to maintain spatial consistency across the dataset. Image intensities were then normalized to the 0-255 range. Using the provided masks, individual intervertebral discs were extracted, and a 5-pixel margin was applied around each disc to define the cropping region. To address class imbalance, additional slices were extracted for underrepresented grades, with five slices per disc for grades 1 and 5, and three slices per disc for grades 2, 3, and 4. Finally, the dataset was split into training and validation sets on a per-patient basis (85% training, 15% validation) to prevent data leakage.

4 Models and methods

We trained several models for degeneration classification:

- Yolo26n-cls
- Custom small CNNs (2 and 4 convolutional)
- ViT-B/16
- 3D CNN

4.1 Yolo26n-cls

The YOLO26n-cls model is a lightweight image classification network consisting of a convolutional backbone and a classification head. The backbone includes multiple convolutional and residual modules that progressively extract hierarchical features from the input image, while the head outputs a single class label for each image. The model has approximately 2.8 million parameters and 0.5 GFLOPs, making it efficient for training and inference. It was pretrained on ImageNet and subsequently fine-tuned on the SPIDER dataset for Pfirrmann grading. During training, different combinations of hyperparameters were tested, including batch size, input image size, number of epochs (with early stopping), and dropout, while various data augmentations such as rotations, flips, color jitter, random erasing, and CutMix were applied.

4.2 Custom-CNNs

We created 2 simple CNNs to get hands-on experience with the topic and check capabilities of the data and models. First version had 3 convolutional layers and the second one had 4 convolutional layers. The input images were resized to 128×128 pixels with three color channels. In the first model, the feature extraction part was composed of three convolutional layers with ReLU activation functions, each followed by a 2×2 max-pooling layer for spatial downsampling. The first convolutional layer used 128 filters with a kernel size of 8, while the subsequent layers used 32 filters with a kernel size of 3. Padding was applied to preserve spatial dimensions before pooling. Second model works similar, however there are 4 convolutional layers instead of 3. The structure of first 3 layers is the same, however the 4th one has the same setting as the 3rd one.

After the convolutional blocks, the resulting feature maps were flattened and passed through a fully connected layer with 100 neurons, followed by the final classification layer producing logits for five output classes.

4.3 ViT transformer model

We used a pretrained ViT-B/16 transformer model with over 16 billion parameters, leveraging transfer learning to adapt the network to our dataset. Pretraining enables the model to benefit from rich, general-purpose visual representations learned from large-scale data, which can significantly improve performance when fine-tuned on smaller medical datasets.

Vision Transformers have become an important tool in computer vision due to their ability to model long-range dependencies and capture global contextual information more effectively than traditional convolutional architectures. Unlike CNNs, which rely on local receptive fields and hierarchical feature extraction, transformers process the entire image as a sequence of patches and apply self-attention mechanisms to learn relationships between distant regions of the image.

From this perspective, the Vision Transformer appeared to be a well-justified and promising choice for this task. Its strong representational capacity and global attention mechanism make it particularly suitable for medical image analysis, where subtle structural patterns and spatial relationships may be crucial for accurate classifica-

tion.

The training process was implemented in PyTorch and conducted with a fixed random seed (42) to ensure reproducibility.

4.4 3D CNN

Unlike the 2D approaches described above, we also experimented with a 3D convolutional neural network to leverage the volumetric nature of MRI data. The rationale for this approach was to capture spatial relationships across adjacent slices, which may contain relevant information for grading disc degeneration. The architecture is 3D CNN with kernel of $3 \times 7 \times 7$.

The preprocessing pipeline for this approach differed from the 2D methods. For each intervertebral disc, a 3D volume was extracted by first projecting the segmentation mask onto 2D, computing the minimum area bounding rectangle using OpenCV, and then extracting and rotating the corresponding region to align the disc horizontally. The extracted volumes were resampled to a fixed size of $224 \times 224 \times 16$ voxels using bicubic interpolation, with z-score intensity normalization applied per volume. Discs for which extraction failed (e.g., due to missing or corrupted mask regions) were excluded from the analysis.

The model consists of an initial 3D convolutional layer with 64 filters (kernel size $3 \times 7 \times 7$), followed by batch normalization, ReLU activation, and max pooling. The feature extraction backbone comprises four residual stages with [2, 2, 2, 2] BasicBlock modules, progressively increasing the channel dimensions from 64 to 512. The model's output are logits for five Pfirrmann grades (1-5). The model contains approximately 33 million parameters. Weights were initialized using Kaiming initialization.

The dataset was split by patient into training (70%), validation (15%), and test (15%) sets to prevent data leakage, with 986 training, 200 validation, and 210 test disc samples from 147, 31, and 32 patients respectively. No external validation set was used. Internal validation was performed on the held-out test partition.

To address class imbalance, weighted cross-entropy loss was employed with weights inversely proportional to class frequencies in the training set. Class counts were: Grade 1 (154), Grade 2 (221), Grade 3 (289), Grade 4 (199), Grade 5 (123).

The model was trained using AdamW optimizer with a learning rate of 10^{-4} , weight decay of 10^{-5} , and batch size of 8 for 50 epochs. The model with highest validation accuracy was selected for final evaluation based on early stopping (13 epochs).

5 Results

5.1 Yolo26n-cls

This model proved to be the most effective for the given problem and dataset, achieving the highest validation accuracy of 67.4%. The model was trained using the following hyperparameter settings:

- a batch size of 32 images
- an input image resolution of 320 pixels
- an initial learning rate of 0.005
- a maximum of 50 training epochs with early stopping enabled after 10 epochs of no improvement (training stopped after 23 epochs, with peak accuracy achieved at epoch 13).

The applied data augmentation techniques included:

- random erasing with a probability of 0.1
- CutMix augmentation with a probability of 0.05
- vertical image flipping with a probability of 0.1
- random rotations of up to 5 degrees
- random scaling and translation of up to 5%.

The confusion matrix (Figure 3) was calculated to provide a detailed breakdown of the model's performance across all five Pfirrmann grades, explaining how the overall accuracy of 67.4% was distributed.

The results show that the model is most effective at identifying advanced stages of degeneration, particularly Grade 3 and Grade 5. Grade 5 achieved the highest reliability, with 163 out of 195 samples correctly identified, while Grade 3 also showed strong results with 137 correct predictions. The high number of correct hits along the main diagonal for these classes suggests that the

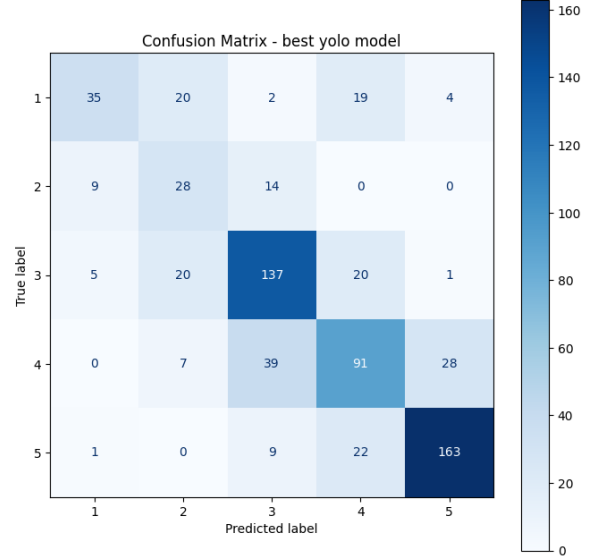


Figure 2: Confusion matrix for best Yolo26n-cls classification.

network successfully learned the visual features of moderate to severe disc degeneration.

Most mistakes occurred between grades that are next to each other on the scale. For example, Grade 4 was mistaken for Grade 3 in 39 cases, and Grade 1 was confused with Grade 2 in 20 instances. These neighboring errors are common in medical grading because the visual differences between consecutive stages can be very subtle. However, a notable anomaly exists where Grade 1 was incorrectly predicted as Grade 4 in 19 cases, representing the largest gap in the model's logic.

The lower accuracy seen in Grade 1 and Grade 2 likely relates to the smaller number of samples available for these categories in the validation set compared to Grade 5.

5.2 CNNs

CNNs were trained on 2 different learning rates (0.001 and 0.0001) and each of them were trained for 20 epochs (however most of them achieved their pick just after a few epochs). Batch size equal to 32 was used. Test accuracy results present as follows:

model	learning rate	test accuracy
CNN (3 layers)	0.001	50.66%
CNN (4 layers)	0.001	54.01%
CNN (3 layers)	0.0001	52.84%
CNN (4 layers)	0.0001	50.51%

Table 1: Accuracies for CNN models

5.3 ViT-B/16

This model was trained on 0.0001 learning rate and batch size 32. It was given 75 training epochs, however it get overfitter, presenting only 44.79% accuracy on a final validation step and 99.11% on a train set. However, during whole training process its accuracy on validation set did not exceed 52% what is a disappointing result taking into consideration popularity of this solution in computer vision

5.4 3D CNN

The 3D CNN model was trained on 986 disc samples and evaluated on a held-out test set of 210 samples. The best validation accuracy of 37.00% was achieved at epoch 13. On the test set, the model achieved an overall accuracy of 48.10% with a macro-averaged F1-score of 0.46 and weighted F1-score of 0.48.

Grade	Precision	Recall	F1-score	Count
1	0.47	0.63	0.54	27
2	0.48	0.62	0.54	63
3	0.48	0.40	0.44	52
4	0.50	0.13	0.21	38
5	0.50	0.63	0.56	30

Table 2: Per-class performance metrics for 3D CNN on test set (n=210).

The results indicate that the 3D CNN struggled most with Grade 4, achieving only 13% recall despite reasonable precision (50%). Grades 1, 2, and 5 showed better recall (62-63%), suggesting the model learned to distinguish the extreme and early-intermediate grades more effectively. Confusion between adjacent grades was common, consistent with the subtle visual differences between consecutive Pfirrmann stages.

The relatively low overall performance compared to the 2D approaches may be attributed to several factors: the smaller effective dataset size when working with full 3D volumes rather than augmented 2D slices, the increased model complexity (33M parameters) requiring more training data to generalize, and potential information loss during the volume extraction and resampling process. The gap between training and validation accuracy also suggests overfitting, which could be mitigated with additional regularization or data augmentation in future work.

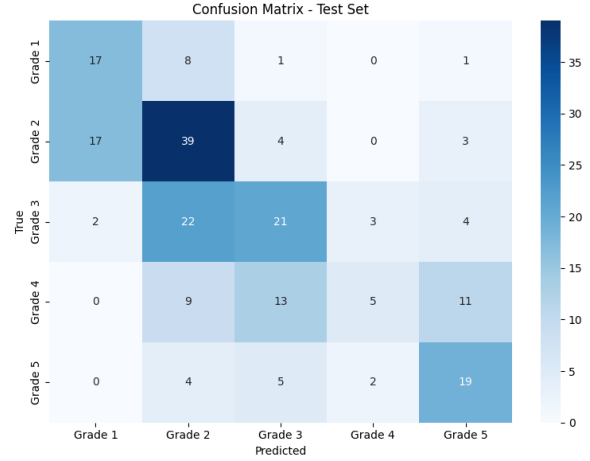


Figure 3: Confusion matrix for 3D CNN model. As we can see the performance is better then it would appear from metrics, as they don't include the ordinality of the classes – the model is often off by 1 (grade 4 being the extreme case).

6 Final pipeline

As part of the project, we developed a final end-to-end pipeline that generates a structured information protocol intended to support clinical decision-making. The implementation, provided in the final.ipynb notebook, enables the selection of a specific patient and the corresponding spine image, based on which an automated diagnostic report is generated.

The pipeline first localizes individual intervertebral discs and it draws bounding boxes around them. Each detected disc region is then cropped and magnified to improve visibility and facilitate further analysis. In addition to localization, the system computes quantitative geometric measurements of each disc, including its depth, width, and height.

Simultaneously, the YOLO26n-cls classification model processes the cropped disc images and assigns a class label to each intervertebral disc. The predicted classifications are integrated into the final report together with the computed measurements and visualizations.

This integrated pipeline combines detection, quantitative analysis, and classification into a unified framework which example result is visible in the figure 4

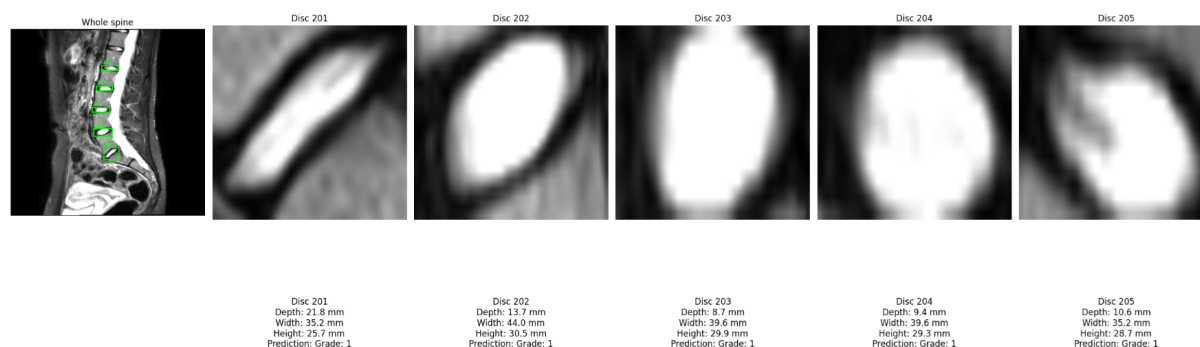


Figure 4: Information table for patient number 237

7 Open science, patient and public involvement

As this was project done for the Artificial Intelligence in Imaging Diagnostics subject conducted on MiNI PW, there was no funding and no conflict of interest. The study protocol was not prepared.

SPIDER dataset is widely known and is available here: [SPIDER dataset link](#). The code is available at our GitHub: [GitHub link](#).

Patients were not involved in the design, conduct, or reporting of this study, as it utilized a publicly available retrospective dataset

8 Discussion

8.1 Interpretation

The YOLO26n-cls model achieved the best performance (67.4% accuracy), outperforming larger architectures including ViT-B/16. This suggests that for limited dataset sizes, lightweight models with appropriate regularization are preferable to high-capacity networks prone to overfitting.

All models showed confusion between adjacent Pfirrmann grades, which is expected given the subtle differences between consecutive classes in Pfirrmann grading. Our results are below those reported in literature.

8.2 Limitations

The main limitations include: (1) small dataset which limits generalizability, (2) class imbalance despite mitigation strategies, (3) overfitting observed in our models, (4) no external validation and (5) absence of confidence intervals for reported metrics.

8.3 Usability of the model in the context of current care

The models require T2-weighted MRI with segmentation masks and technical expertise to run. For clinical use, integration into radiology workflows would be needed. The predictions should serve as decision support rather than replace radiologist judgment. Future work should focus on external validation, uncertainty quantification, and explainability methods to increase clinical trust.

9 TRIPOD+AI checklist

- Title meets criteria (item 1)
- Abstract topic is not applicable (item 2)
- Introduction items are specified in section 1 (items 3 and 4)
- Dataset specific informations are in section 2 (items 5, 6, 10, 20)
- Data processing informations are in section 3 (items 7, 9, 11, 13, 16, 21)
- Model specific information is in section 4 (items 12, 15, 22, 23)
- Item 8 is inherently answered by the title and whole raport
- Items 18 and 19 are answered in section 7
- Interpretation, limitations and Usability of the model in the context of current care are answered in sections 5 and section 8
- Not applicable items: 14, 17, 24

References

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